

Myeloid Neoplasms - Factoid Sheet

Week 7 Hematology • Lecture 8 • Savage • memorize-first companion

ScholarRx: [Chronic Myeloproliferative Disorders](#) • [AML](#) • [CML](#) • [PV](#)

Core framework: AML = immature blasts • MDS = ineffective/dysplastic production • MPN = too many mature cells • MDS/MPN = dysplasia + proliferation.

AML / APL

Factoid	Memorize
AML	Usually >=20% myeloid blasts unless a defining genetic lesion allows diagnosis below threshold.
Auer rod	Definitive myeloid lineage clue.
Marrow failure	Anemia -> fatigue; thrombocytopenia -> bleeding; neutropenia -> infection.
t(8;21)	Favorable recurrent AML association in lecture.
inv(16)	Favorable; abnormal eosinophils / myelomonocytic disease.
APL	t(15;17) PML-RARA + abnormal promyelocytes/Auer rods + DIC.
APL action	Start ATRA immediately when suspected - do not wait for FISH.
Differentiation syndrome	Fever, edema/weight gain, dyspnea/pulmonary edema, hypotension, rising WBC -> dexamethasone.

MDS

- Hypercellular marrow + peripheral cytopenias
- Ineffective hematopoiesis / dysplasia
- Pseudo-Pelger-Huet neutrophils
- Micromegakaryocytes
- Ring sideroblasts
- Can progress to AML

Classification trap

The official LO is conceptual. Savage marks detailed MDS WHO/ICC tables “**No need to memorize.**” Learn recurring fingerprints, not every category.

CML

t(9;22) Philadelphia -> BCR-ABL1 constitutive tyrosine kinase -> leukocytosis + full myeloid left shift + basophilia + low LAP -> TKI.

Clue	Meaning
Basophilia	Strongly favors CML over simple leukemoid reaction.
Low LAP	CML; reactive leukemoid response tends to have high LAP.
FISH/karyotype	Detect BCR-ABL1 / Philadelphia chromosome.
PCR	Quantitative molecular monitoring after diagnosis.
Imatinib	Binds BCR-ABL kinase and blocks phosphorylation/downstream signaling.

Classic MPN Factoids

PV vs ET vs PMF

	PV	ET	PMF
Dominant	RBC / panmyelosis	Platelets	Megakaryocytes + granulocytes -> fibrosis
Genes	JAK2	JAK2/CALR/MPL	JAK2/CALR/MPL
Blood	High Hgb/Hct; EPO low	Persistent thrombocytosis	Leukoerythroblastic smear; cytopenias later
Morphology	Panmyelosis	Large hyperlobulated "staghorn" megakaryocytes	Teardrops + marrow fibrosis + dry tap
Clinical	Aquagenic pruritus, erythromelalgia, thrombosis	Thrombosis OR bleeding	Massive splenomegaly / extramedullary hematopoiesis

PV clues • Low EPO = autonomous production • Secondary erythrocytosis tends to have high EPO • Hot-shower itching • Hyperviscosity / thrombosis • Erythromelalgia	ET paradox • Many platelets can clot • But extreme thrombocytosis can cause acquired vWD and bleeding • Rule out reactive thrombocytosis
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PMF mechanism Abnormal megakaryocytes release fibroblast-stimulating cytokines -> reactive marrow fibrosis -> dry tap + myelophthisic smear + extramedullary hematopoiesis -> huge spleen.

Low-priority CMML gap-fill

Criterion	Companion recognition rule
Monocytes	Absolute $\geq 0.5 \times 10^9/L$ AND $\geq 10\%$ of WBC
Blasts	$< 20\%$
BCR-ABL1	Absent
Morphology	Dysplasia in at least one lineage

If you see X -> think Y

• Multiple Auer rods + bleeding/DIC -> APL • Busy marrow + cytopenias + pseudo-Pelger -> MDS • WBC 150K + basophilia + myelocytes -> CML • High Hct + low EPO + itching after shower -> PV	• Platelets $> 450K$ + staghorn megakaryocytes -> ET • Teardrops + dry tap + massive spleen -> PMF • Persistent monocytosis + dysplasia + BCR-ABL1 negative -> CMML
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Last 5 minutes: APL=15;17 PML-RARA/DIC/ATRA. CML=9;22 BCR-ABL1/basophilia/low LAP/TKI. PV=JAK2/high Hct/low EPO. ET=JAK2-CALR-MPL/platelets. PMF=JAK2-CALR-MPL/teardrop/dry tap/spleen.

Primary source: Savage 44-slide Myeloid Neoplasms deck. Official Week 7 objectives control emphasis. The uploaded companion is used for broad treatment logic, CMML, and board-style distinctions.